

In this Standard the comma has been adopted as the decimal sign.

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1. SCOPE

This Standard establishes general tolerances for metal sheet or extruded parts.

2. APPLICATION

This Standard shall be applied, whenever it is called out on drawing (see Item 6.1), for dimensions or geometric configurations without specific tolerances or if there are no tolerances according to other mandatory standards called out on the drawing too.

NOTE

If the tolerances established in this Standard do not assure the function or interchangeability of the part, specific tolerances shall be indicated on drawing.

3. APPLICABLE DOCUMENTS

Embraer Standards

NE 03-006 - Rivets and Pins - Drilling, Countersinking and Dimpling.

NE 03-051 - Metallic Parts - Cutting, Chamfering and Deburring.

NE 03-071 - Machined Metallic Details - General Tolerances.

NE 03-073 - Dimensioning and Tolerancing.

NE 03-074 - Dimensional Tolerances for Aluminum Mill Products.

NE 03-075 - System of Limits and Fits - ISO 286.

NE 40-006 - Aluminum and its Alloys - Chemical Conversion Coating.

NE 40-261 - Aluminum Clad Sheet Surface Skin Quality - Surface Acceptance Criteria.

NE 41-013 - Aluminum and its Alloys - Chemical Milling.

4. DEFINITIONS

4.1 GENERAL TOLERANCE

Common tolerance to several dimensions or geometric configurations. It is not indicated next to the nominal dimension on the drawing.

NOTE

Specific tolerances (see Item 4.2), indicated directly on the drawing or through the call of another specific standards, are prioritized with relationship to general tolerances.

4.2 SPECIFIC TOLERANCE

Tolerance indicated next to the nominal dimension or configuration on drawing (see note of Item 4.1).

4.3 ORANGE PEEL

Surface roughness or waving with rough or granular characteristics.

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4.4 SURFACE CLASS

Surface characteristics are required so that the part to which the surface belongs may perform its function adequately.

The surfaces are classified into four classes: I, II, III and IV (see Table 2).

4.5 WARPING

Concomitant geometrical deformations (bend torsion) of a formed part.

4.6 MODEL BASED DEFINITION (MBD)

Illustrative 3D drawing that captures every detail of product, project information and engineering requirements.

4.7 SHEET METAL PARTS

Sheet metal parts are metallic aircraft parts made from a metallic sheet metal (thickness up to 6,35 mm), composed basically by a main body (web) with one or more flanges in their edges. Drilled holes, flanged holes or flanged cut-outs may be present on the web or in the flanges.

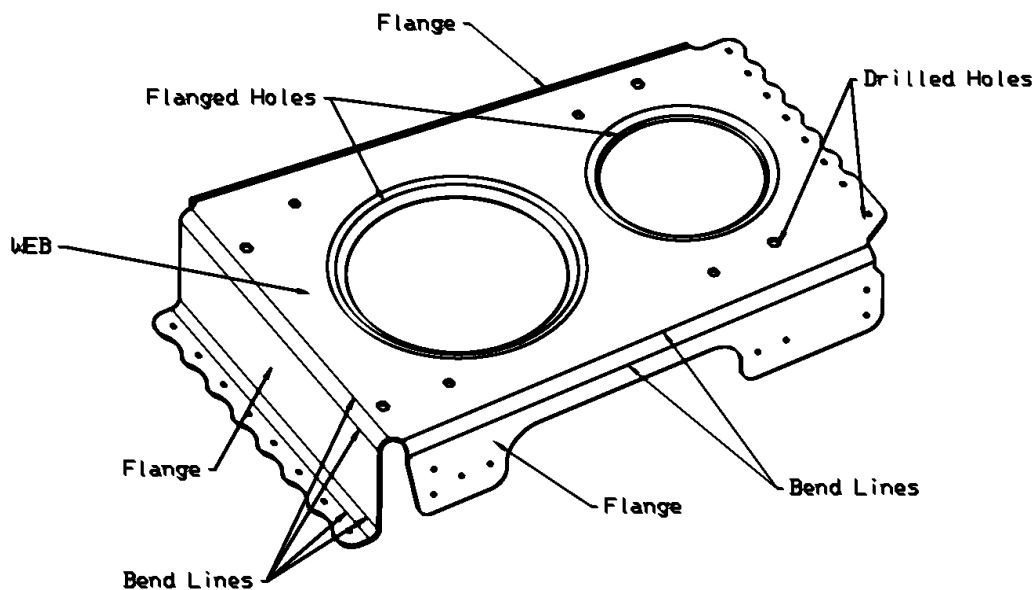


FIGURE 1 : SHEET METAL PART MAIN DETAILS

4.8 EXTRUDED PARTS

Parts made from a metallic extruded shape (Z, T, L, etc.) are composed basically by a main body (web) with one or more legs.

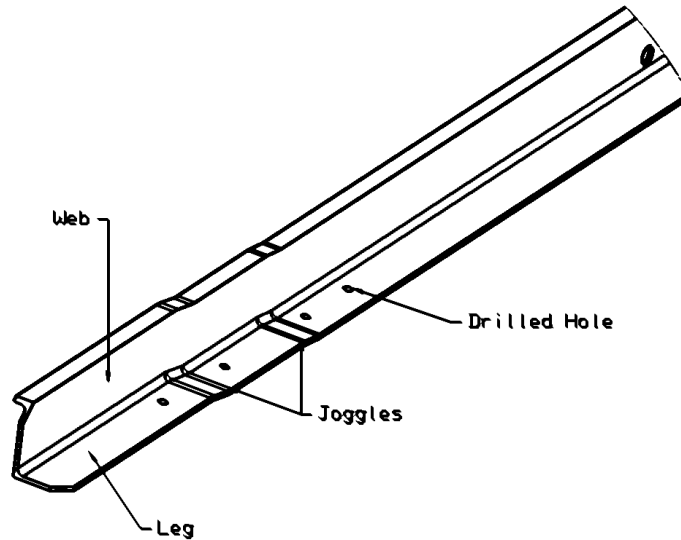


FIGURE 2 : EXAMPLE OF AN AIRCRAFT PART MADE FROM Z SHAPE

5. REQUIREMENTS

5.1 DIMENSIONAL TOLERANCES

5.1.1 Linear and Angular General Tolerances

Tolerances for linear and angular dimensions shall be according to Figure 3 and Table 1.

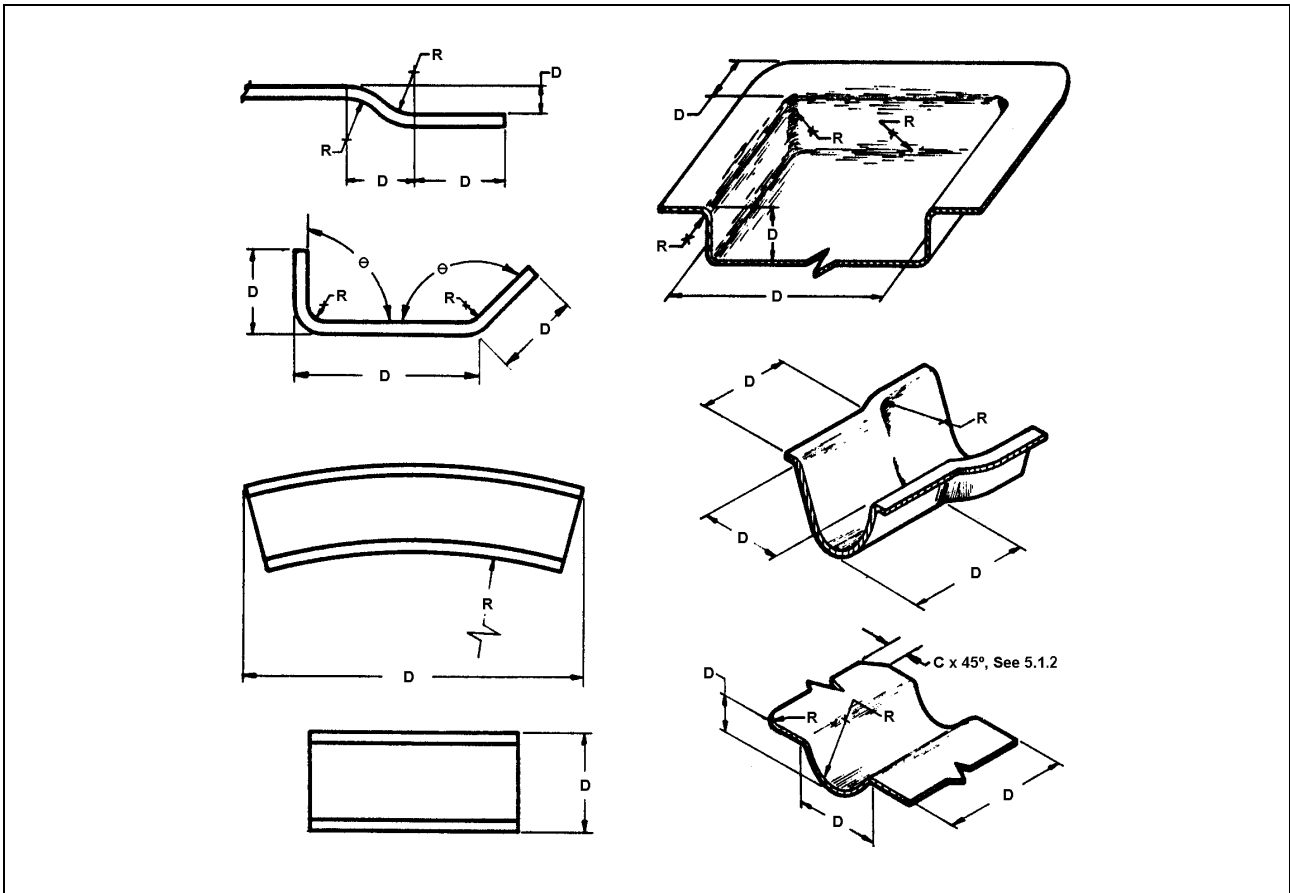


FIGURE 3 : TOLERANCES FOR LINEAR AND ANGULAR GENERAL DIMENSIONS

TABLE 1 : TOLERANCES FOR LINEAR AND ANGULAR GENERAL DIMENSIONS

NOMINAL LINEAR DIMENSION "D" OR "R" (mm)	TOLERANCE $\pm$ (mm) (1)	NOMINAL ANGULAR DIMENSION (INCLUDING 90° NOT QUOTED)	TOLERANCE $\pm^\circ$
Over 0 up to 300	0,8	Over 0° up to 360°	(2)
Over 300 up to 1000	1		
Over 1000	0,0012D (or R)		

(1) Exceptions:

- The general tolerance for bending radii $R < 50$ mm in cold bending of metal sheets shall be on the plus side only. Example: $R = 30$ mm varies from 30,0 mm to 30,8 mm.
- The deformation tolerance for parts with all surface class IV (see Item 5.2) shall be ± 2 mm.
- The general tolerance of flange height of parts bent on hydroforming press or rubber press shall be $+1,4$ mm / $-1,0$ mm.

(2) Consult Item General Angle Variations of Table 6.

NOTE

For MBD methodology, dimensions not shown in the production drawing shall be measured directly in the 3D model and for not tolerated geometries, apply the general tolerances herein specified. For the manufacturing, the dimensions shall be informed by the operations outline.

NOT APPLICABLE FOR NEW DRAWINGS AFTER 2023/May/10.

5.1.1.1 General Tolerances for Holes

The general tolerances for holes shall follow the same requirements stated in NE 03-071, with exception of the information about tolerance for hole for rivet installation (this type of drilling is called on drawings thru respective NIs).

5.1.2 Drilled Holes General Tolerances**5.1.2.1 Holes for Bolt Installation**

The pitch tolerance for bolt installation holes shall be according to the specific tolerance indicated on drawing.

The distance from edge tolerance for bolt installation holes shall be according to Table 2.

TABLE 2 : EDGE TOLERANCE FOR BOLT INSTALLATION

NOMINAL LINEAR DIMENSION "D"	TOLERANCE (mm)
$D \leq 1,5$	+0,3 -0,1
$1,5 < D \leq 3$	$\pm 0,3$
$3 < D \leq 6$	$\pm 0,4$
$6 < D \leq 30$	$\pm 0,5$
$30 < D \leq 100$	$\pm 0,8$
$100 < D \leq 2000$	$\pm 1,0$
$2000 < D$	$\pm 0,0006D$

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5.1.2.2 Hole Diameter

Tolerance for hole diameter shall be according to Table 3.

TABLE 3 : GENERAL TOLERANCES FOR ROUND HOLES

NOMINAL HOLE DIAMETER Ø (mm) (1)	TOLERANCE OF Ø (mm) (3)
$- < \varnothing \leq 3$	+0,25 0
$3 < \varnothing \leq 6$	+0,30 0
$6 < \varnothing \leq 10$	+0,36 0
$10 < \varnothing \leq 18$	+0,43 0
$18 < \varnothing \leq 30$	+0,52 0
$30 < \varnothing \leq 50$ (2)	+0,62 0

(1) Centering holes (guide holes) dimensioned with Ø2,1 mm on drawings may be drilled (preferably) with Ø2,5 mm since the manufacturing cost justifies the alteration.

(2) Hole diameters larger than 50 mm shall have tolerances according to H14 of NE 03-075.

(3) Hole diameter for fasteners installation:

- hole diameters for bolt/screw installation shall be according to specific tolerance indicated on drawing;

- hole diameter for rivet installation shall be according to NE 03-006 and according to specified on drawing; and

- 2,5 mm diameter guide holes in curved flanges of sheet metal primary parts, which will undergo subsequent enlargement for fastener installation in assembly line, shall be approved with diametric variation of +0,5 mm/-1,0 mm. These holes shall not be used as coordinate holes.

5.1.2.3 Hole Center Line

The tolerance for angular displacement of hole center line is $\pm 1^\circ$.

The tolerance for linear displacement of hole center line is ± 1 mm.

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5.1.2.4 Distances Between Holes Located in Flat Web of Straight Contour Flanged Sheet Metal Parts or not Flanged Flat Sheet Metal Parts

The tolerances shown in Table 4 are valid for a group of “N” drilled holes, (except tooling holes and guide holes) with diameter ranging from 2,5 mm to 9,1 mm and bend line distance not smaller than 7 mm, located in flat web of straight contour flanged sheet metal parts (see Figure 4) or in not flanged flat sheet metal parts (see Figure 5). The premises established on 5.1.2.5 are conditioned only to the limitations and characteristics of the manufacturing process.

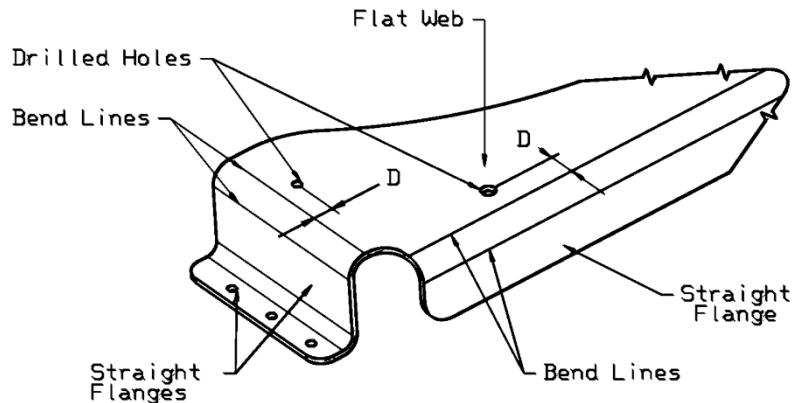


FIGURE 4 : FLAT WEB OF STRAIGHT CONTOUR FLANGED SHEET METAL PART

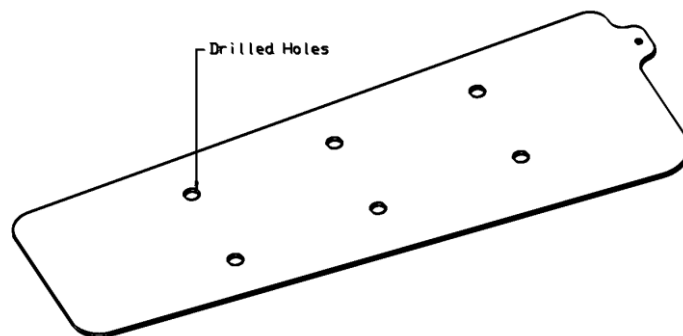


FIGURE 5 : NOT FLANGED FLAT SHEET METAL PART

TABLE 4 : TOLERANCES FOR DISTANCE BETWEEN HOLES

DISTANCE BETWEEN HOLES "D" (mm)	LINEAR TOLERANCE (mm)	POSITION TOLERANCE (mm) REF (1) (2) (3)
$D \leq 140$	$\pm 0,15$	$\Phi \quad \Phi 0,4 \text{ (M)}$
$140 < D \leq 400$	$\pm 0,20$	$\Phi \quad \Phi 0,6 \text{ (M)}$

(1) Position tolerance equivalence informed just as a reference. Production shall use the range of linear tolerance.

(2) For groups of only 2 holes, use the range of the linear tolerance.

(3) See NE 03-073.

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5.1.2.5 General Tolerances for Pitch and Edge Distance

The general tolerances shall follow the requirements stated in NE 03-006 except for bolt installation (see Item 5.1.2.1).

5.1.3 General Tolerances for Distances Between Cutouts on Instrument Panels

Table 5 shows the tolerances for distances between cutouts on instrument panels.

TABLE 5 : TOLERANCES FOR DISTANCES BETWEEN CUTOUTS ON INSTRUMENT PANELS

NOMINAL LINEAR DIMENSION "D" (mm)	TOLERANCE (mm)
$D \leq 1,5$	+0,3 -0,1
$1,5 < D \leq 100$	$\pm 0,3$
$100 < D \leq 400$	$\pm 0,4$
$400 < D \leq 1000$	$\pm 0,6$
$1000 < D \leq 2000$	$\pm 0,8$
$2000 < D$	$\pm 0,0006D$

NOTE

For machined details such as sharp edges, radio, chamfers and others (see NE 03-071).

5.1.4 Shop Corner Options of Sheet or Extruded Parts

If drawing does not indicate the note "NL XXX - CORNER SHALL BE AS SHOWN", shop may change corners according to options specified in Figure 6.

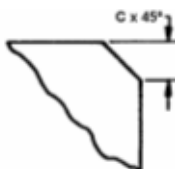
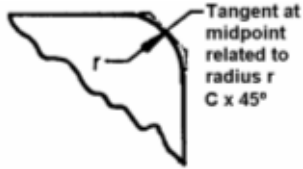
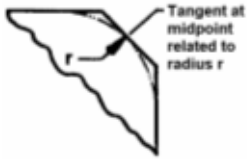
SHOWN ON DRAWING	SHOP OPTION(S)	
		
		
		
		

FIGURE 6 : SHOP CORNER OPTIONS

5.1.5 Radii Relief

Although other ways specified on drawing, sheet parts containing corners formed by two convergent flanges presenting problems upon forming such as bending radius smaller than the minimum specified value, cracks, superficial deformations, etc. shall present a radius relief according to one of the configurations specified in Figure 7.

NOTES

- 1) Radius relief shall not interfere in the functional performance of part (reduction of edge distance for rivet installation, reduction of mechanical strength, etc.). In case of doubt, consult Structure Engineering.
- 2) Class IV parts according to Item 5.2 do not require radius relief.

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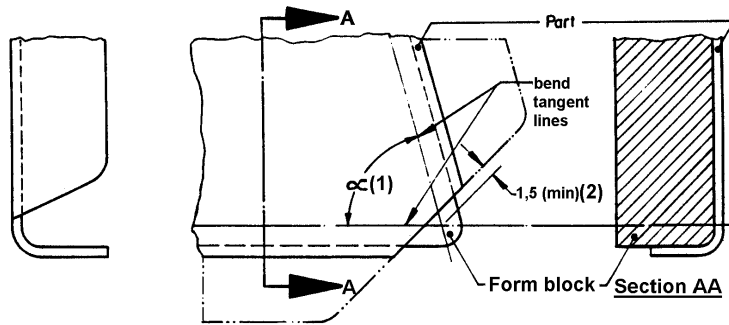
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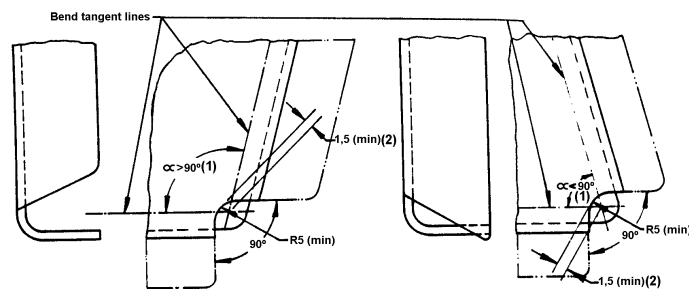
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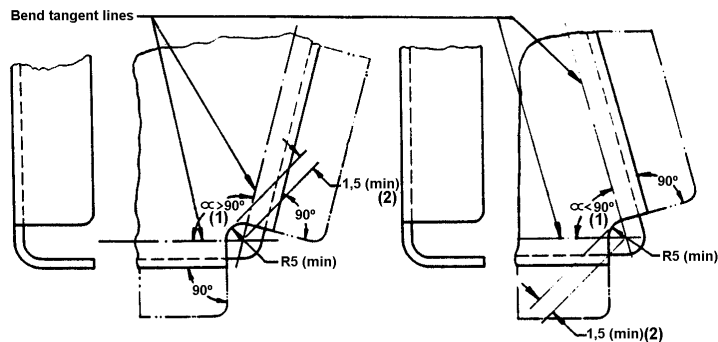
1st CONFIGURATION: STRAIGHT CUT OUT ON BLANK



NOTE

It shall be considered, in this case, the final position of bent flange if the angle α is different from 90° .

2nd CONFIGURATION: CUT OUT UNDER ANGLE 90° ON BLANK



NOTE

This cut out promotes flange corners with 90° , independent of angle α .

3rd CONFIGURATION: PERPENDICULAR CUT OUT TO THE BEND TANGENT LINES ON BLANK

(1) Angle α may be equal or different from 90° for all three configurations.

(2) The maximum value ($\approx 2,5$ mm) is free, but it shall not interfere on functional performance of part (edge distance, etc.).

FIGURE 7 : RADIUS RELIEF

5.1.6 Lugs Cutout

5.1.6.1 General Rule

The symbol  in lug representation on drawings means that it shall be cutout according to part shape (see Figure 8).

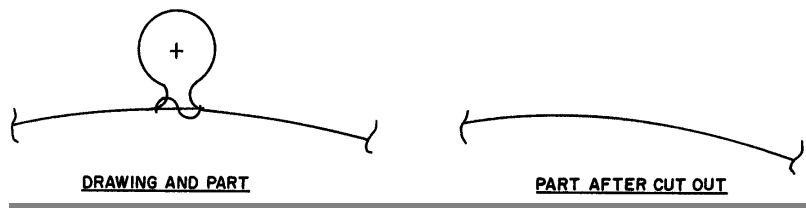


FIGURE 8 : LUG CUTOUT GENERAL RULE

5.1.6 Lugs Cutout

5.1.6.1 General Rule

The symbol  in lug representation on drawings means that it shall be cutout according to part shape (see Figure 8).

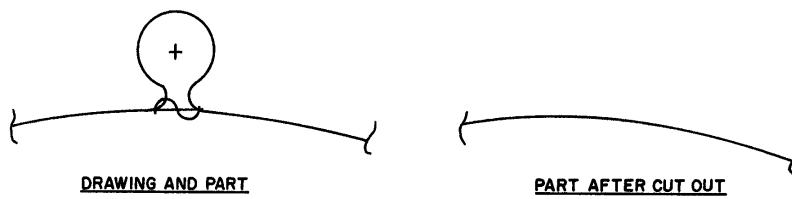


FIGURE 8 : LUG CUTOUT GENERAL RULE

5.1.6.2 Lug on Winding Shape

In this case the lug cutout shall be always straight, according to indicated on Figure 9.

The part is authorized to have different contour from that indicated on drawing, in the lug (cutout) region.

a) Lug Indicated on Drawing

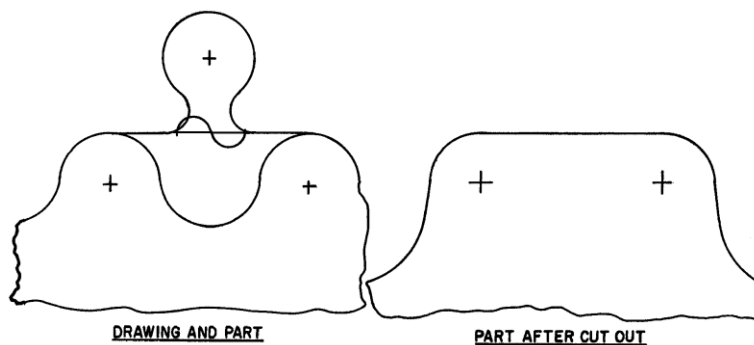
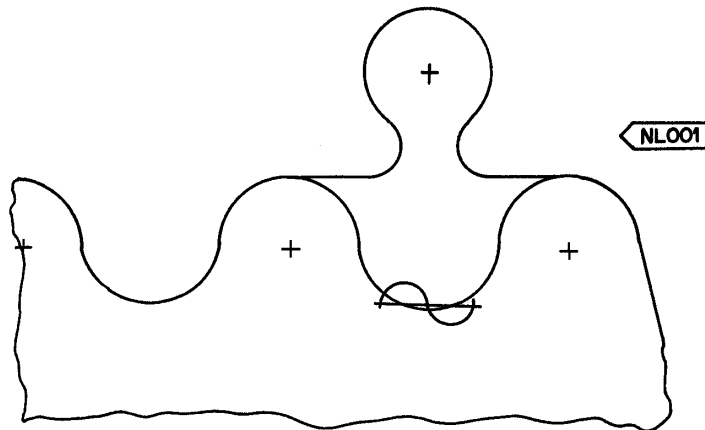


FIGURE 9 : LUG INDICATED ON DRAWING

NOTE

If the winding shape shall remain after a lug cutout, a free note (NL XXX) shall be added on drawing. See example in Figure 10.



NL 001 - The winding shape shall remain after lug cutout.

FIGURE 10 : EXAMPLE OF FREE NOTE INDICATION FOR WINDING SHAPE REMAIN

b) Lug Not Indicated on Drawing

In this case the lug is only made to facilitate the manufacturing process and is added to the part according to production necessity. See example in Figure 11.

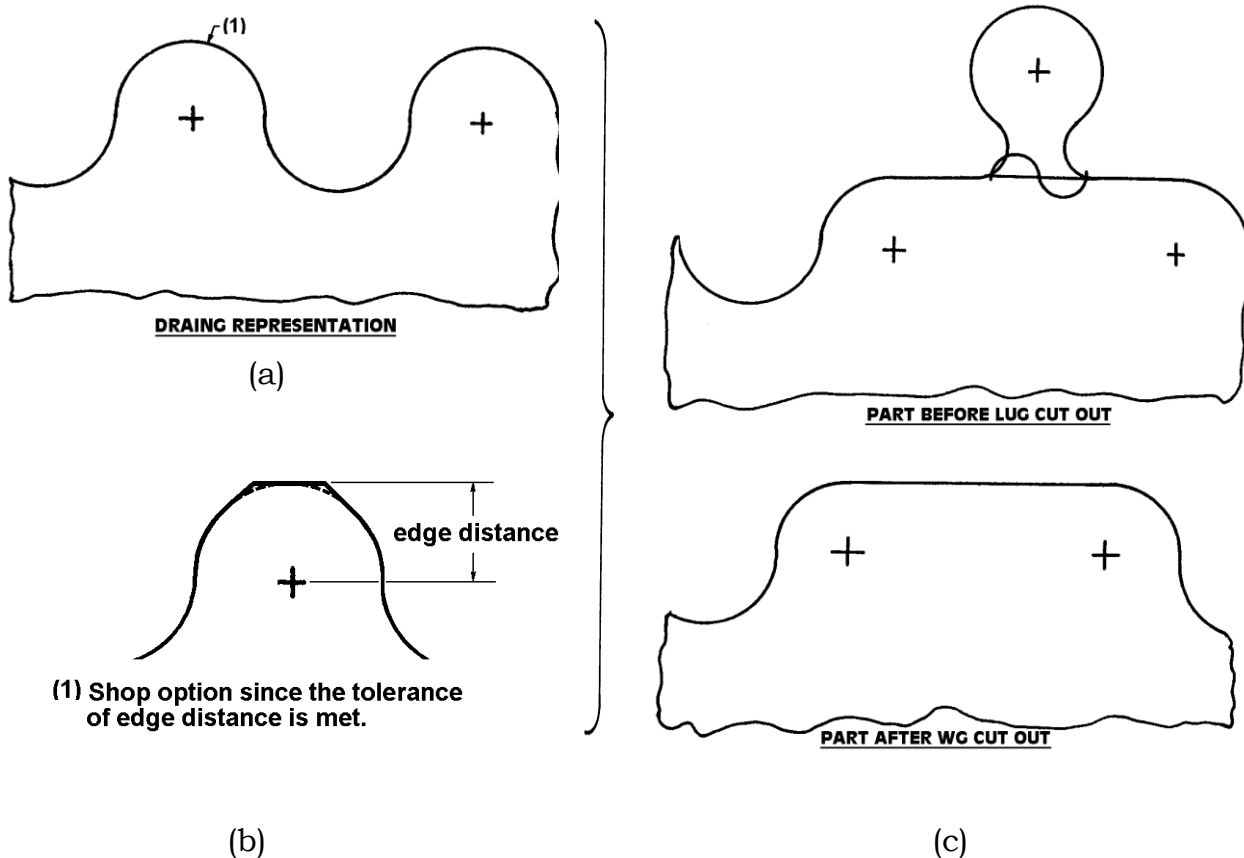


FIGURE 11 : LUG NOT INDICATED ON DRAWING

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5.1.7 Thickness Reduction Induced by Sheet Forming

Not applicable for new drawings.

~~The maximum reduction allowed is 9% for parts Class I, 17% for Classes II and III and without control for Class IV.~~

For new drawings the thickness reduction shall be classified through Thickness Reduction grades according to as follows:

1. GRADE 9 - Thickness reduction up to 9% after the forming process.
2. GRADE 13 - Thickness reduction up to 13% after the forming process.
3. GRADE 17 - Thickness reduction up to 17% after the forming process.
4. GRADE 23 - Thickness reduction up to 23% after the forming process.
5. GRADE 30 - Thickness reduction up to 30% after the forming process.
6. GRADE 40 - Thickness reduction up to 40% after the forming process.

This reduction shall be determined considering the thickness before and after the last forming step and shall be calculated as follows:

$$\% \text{ Thickness Reduction} = \left[\frac{\text{Thickness(initial)} - \text{Thickness(final)}}{\text{Thickness (initial)}} \right] \times 100$$

Where:

Thickness (initial), is the sheet thickness measured before the forming process.

Thickness (final), is the sheet thickness measured at formed area (area with plastic deformation).

NOTE

Whenever no thickness reduction control and inspection are required EN 9014 shall be called out on drawing (see Item 6.1.2).

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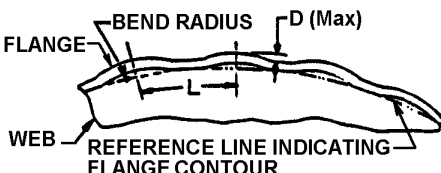
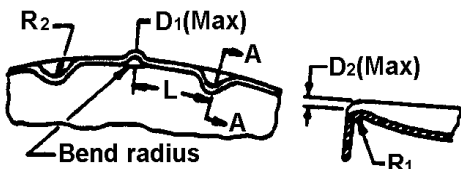
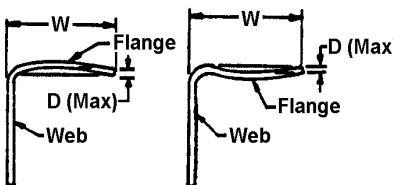
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5.2 SURFACE DEFORMATION TOLERANCES

5.2.1 Class Tolerances

They shall be within the limits specified in Table 6.

TABLE 6 : CLASS TOLERANCES

SURFACE DEFORMATION		TOLERANCES (mm)				
		CLASS I (3)	CLASS II (4)	CLASS III (5)	CLASS IV (6) (1)	
SURFACE WARPING 		D(Max) = 0,8		D(Max) = 1,5		
		R ≥ 5 x t				
WRINKLES IN CONVEX FLANGES 		D(Max) = 0,8		D(Max) = 1,5		
NOTES 1) W = Flange width 2) Measurements shall be taken perpendicular to the flange from adjacent through to enclosed peak.		L ≥ W (2)				
FLUTED CONVEX FLANGES 		D1(Max) = 0,8		D1(Max) = 1,5	D1(Max) = 2,6	
		D2(Max) = 1,3				
		L ≥ W (2)				
		R1 and R2 ≥ R				
FLANGE CAPPING 		W (mm)	D(Max) =	(Max) =	D(Max) =	D(Max) =
		10 < W ≤ 20	0,12	0,22	0,90	1,50
		20 < W ≤ 30	0,20	0,38	1,50	2,50
		30 < W ≤ 40	0,28	0,52	2,10	3,50
		40 < W	0,008xW	0,015xW	0,06xW	0,1xW

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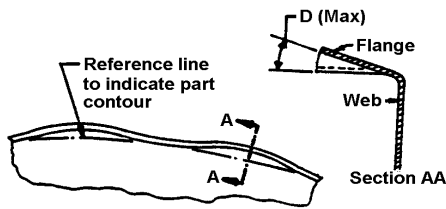
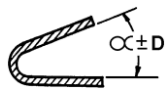
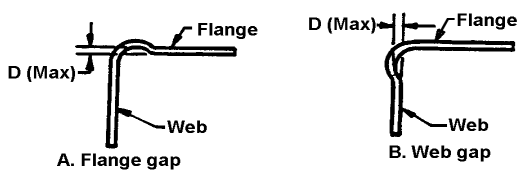
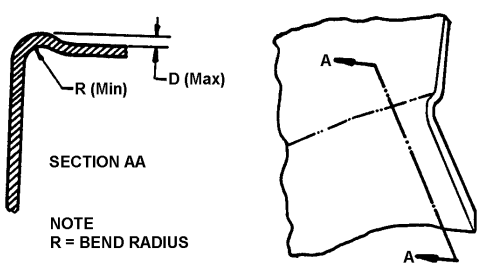
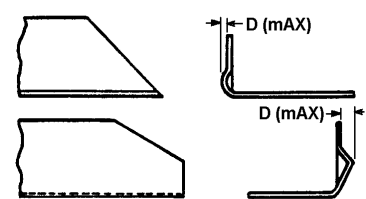
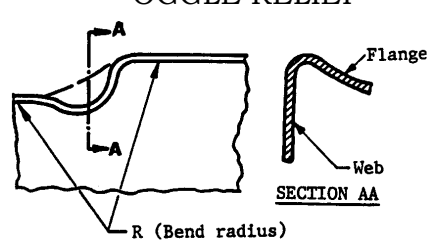
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TABLE 6 : CLASS TOLERANCES

SURFACE DEFORMATION	TOLERANCES (mm)			
	CLASS I (3)	CLASS II (4)	CLASS III (5)	CLASS IV (6) (1)
LOCALIZED ANGLE VARIATIONS OF FLANGE BEND 	$D(\text{Max}) = 2^\circ$		$D(\text{Max}) = 6^\circ$	$D(\text{Max}) = 10^\circ$
GENERAL ANGLE VARIATIONS EXCEPT FOR THE LOCALIZED ONES 	$\pm D = \pm 1^\circ$	$\pm D = \pm 2^\circ$	$\pm D = \pm 2^\circ:30'$	$\pm D = \pm 5^\circ$
GAPS 	$D(\text{Max}) = 0,3$	$D(\text{Max}) = 0,8$	$D(\text{Max}) = 1$	$D(\text{Max}) = 2,6$
GRIMPING OF THE WEB ADJACENT TO JOGGLE FLANGE 	$D(\text{Max}) = 0,3$	$D(\text{Max}) = 0,8$	AS MANUFACTURED	
	$R(\text{Min}) = R - 0,25R$			
FLANGE EDGE BULGING 	$D(\text{Max}) = 0,5$	$D(\text{Max}) = 0,8$	AS MANUFACTURED	
OGGLE RELIEF 	THERE SHALL NOT BE JOGGLE RELIEF			

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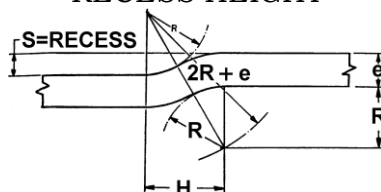
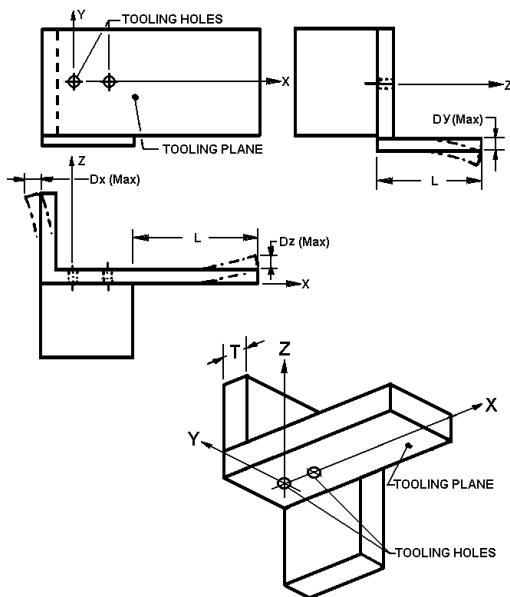
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TABLE 6 : CLASS TOLERANCES

SURFACE DEFORMATION	TOLERANCES (mm)			
	CLASS I (3)	CLASS II (4)	CLASS III (5)	CLASS IV (6) (1)
RECESS HEIGHT 	H ±0,3	H ±0,5	H ±0,8	
WARPING 	TOTAL WARPING Dx(max) = Dy(max) = Dz(max) = 0,8 mm for total warping.			
USE OF WEIGHTS If the warping on first measurement is larger than the tolerance value, the measurement shall be repeated using weights as follows: a) 0,5 kgf for thickness (T) ≤1; and b) 1 kgf for thickness (T) >1. The weights shall be applied on the warping detail on two perpendicular arranged directions, at 250 mm intervals (one weight to the other one) or proportionally for smaller intervals. Example: half of the specified weight at 125 mm intervals. There are two tolerances for weight application: a) 0,0 (zero) for places just under the weight location; and b) the warping general tolerance (0,8 mm), used for measurement, without weight application, for the other places. NOTES 1) Before warping measurement, the part shall be supported by a device that represents its assembling on the aircraft (it shall be used the parts datums, such as tooling holes, tooling planes or other important details for the part assembling). 2) See Item 6.1.8.				

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(1) a) The requirements specified in the Item 5.2 of this Standard refer to the part surfaces or profile. They shall not be applied to the bending line, dimensions and tolerances of bending lines nor to the inspection template.

b) The inspection of Class IV surface deformations after the manufacturing of parts is not necessary. The tolerances specified in this Standard for Class IV are only for reference to manufacturing and tooling. Other information called on drawings (X-ray, penetrant liquid, etc.) shall be performed normally.

(2) This allowance is only applicable to gradual changes in dimension such as bow or waviness and shall not be construed as allowing abrupt, localized distortions such as kinks, creases, etc.

(3) Flanges which attach to or affect the internal or external lofted contour of the wing, control surfaces, nacelles and pylons. For $t > 0,040$ inch, the use of this Class is mandatory.

(4) Flanges which attach to or affect the internal or external lofted contour of fuselage. Used only with $t \leq 0,040$ inch.

(5) Flanges that mate with or attach internally to other parts.

(6) Free flanges (used for stiffener purposes only).

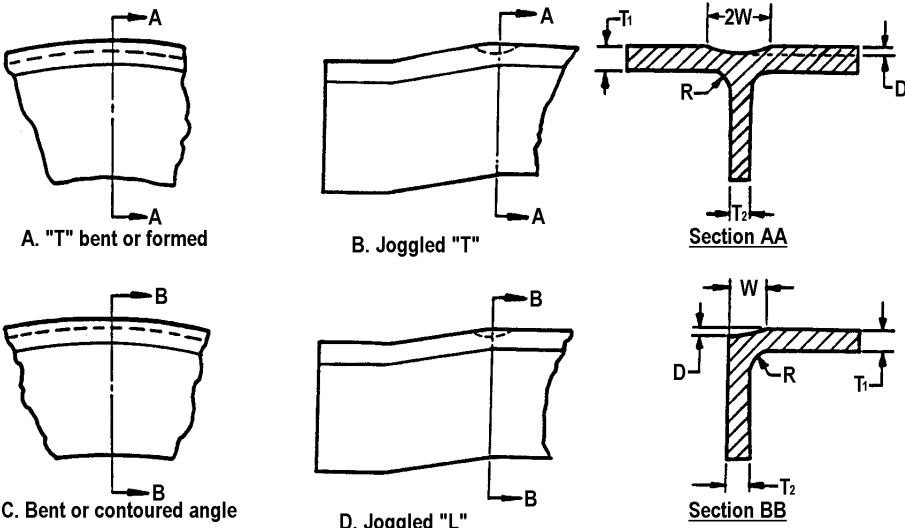
5.2.2 Tolerances Regardless of Classes

The tolerances specified herein shall be applied regardless of the surface class.

5.2.2.1 Depressions of the Leg Area

They shall be within the limits specified in Table 7.

TABLE 7 : DEPRESSIONS OF THE LEG AREA

			
PROFILE	T_1 (mm)	MAXIMUM DEPRESSION (mm)	
		D	W
T	$T_1 < 3,2$	$0,2 \times T_1$	$2R + T_2$
	$3,2 \leq T_1$	0,7	
L	$T_1 < 3,15$	$0,2 \times T_1$	
	$3,2 \leq T_1$	0,7	

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5.2.2.2 Joggle Corner Relief Radius

Corner relief radius shall not reach the center line of the adjacent holes for the rivets (see Figure 12).

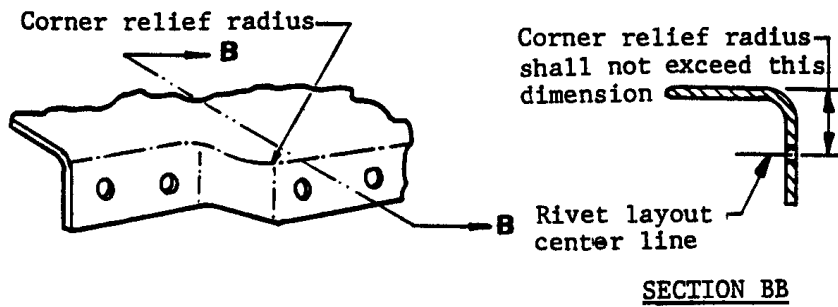


FIGURE 12 : TOLERANCE FOR JOGGLE CORNER RELIEF RADIUS

5.2.2.3 Cracks, Orange Peel Type Defect Luder Lines and Scratches

- a) Parts presenting cracks shall be rejected.
- b) Parts presenting orange peel type defect shall be submitted to Materials Engineering for evaluation.
- c) Formed parts showing Luders lines crossing the surface shall be rejected for final polished surface. For painted parts these defects are acceptable providing they are not visible after final top coating.
- d) Scratches that may be noticeably felt with fingernails on the sheet surfaces shall be removed regardless of the part's tolerance class. After removing the scratch apply a chemical conversion coating according to NE 40-006, Type III, on areas where the clad has been removed. The maximum surface roughness (Ra) shall be 3,2 μm and the remaining thickness shall comply with the dimensional tolerance specified in NE 41-013 for regions within of the chemical milling area. For regions outside, the requirements of NE 03-074 for aluminum alloys shall apply. For other materials, the remaining thickness shall meet the tolerances defined in their respective Material Specification documents otherwise specified in the drawing.
- e) Oxide streaks which were accepted during receiving inspection (as defined in NE 40-261), based on material specification, shall be removed. The maximum final roughness (Ra) shall be 3,2 μm . For clad sheets, assure the sanding process does not go through the clad layer exposing the core of the sheet. In case the core is exposed, apply chemical conversion coating according to NE 40-006, Type III, where the clad layer was removed. The remaining thickness shall comply with the dimensional tolerances specified in NE 41-013 for regions within the chemical milling area. For regions outside the chemical milling area, the requirements of NE 03-074 for aluminum alloys shall apply. For other materials, the remaining thickness shall meet the tolerances defined in their respective Material Specification documents, unless otherwise specified in the drawing.

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5.2.2.4 Surface Finish

- a) Parts cut by pressing or machining shall present surface roughness of $Ra = 3,2 \mu m$.
- b) Parts shall be deburred according to NE 03-051.

5.3 GRAIN FLOW DIRECTION

Grain flow direction tolerance (general or specific) is applicable to sheet parts.
Consider the following cases:

- a) drawing does not indicate grain direction but calls out NI 383 (refer to Item 6.1.7). In this case, grain direction is optional and may be established according to manufacturing convenience;
- b) drawing indicates grain direction but does not indicate specific angular tolerance. In this case, general angular tolerance to be considered shall be $\pm 15^\circ$;

EXAMPLE:

grain direction

- c) drawing indicates grain direction and a specific angular tolerance. In this case, indicated tolerance is mandatory.

EXAMPLE:

grain direction $\pm 10^\circ$

6. GENERAL NOTES

6.1 DRAWING CALLOUT

Not applicable for new drawings.

~~6.1.1 This Standard shall be called out on the production drawings through the following Item Notes (NI), as applicable:~~

- ~~a) "NI 973 - General tolerances for sheet or extruded parts according to NE 03-080, class I";~~
- ~~b) "NI 974 - General tolerances for sheet or extruded parts according to NE 03-080, class II";~~
- ~~c) "NI 975 - General tolerances for sheet or extruded parts according to NE 03-080, class III";~~
- ~~d) "NI 976 - General tolerances for sheet or extruded parts according to NE 03-080, class IV";~~

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6.1.1 For new drawings, the geometric tolerance classes shall be called on drawing through the following Notes (EN), as applicable:

- a) EN 9027 - General tolerances for sheet or extruded parts according to NE 03-080, Class I;
- b) EN 9028 - General tolerances for sheet or extruded parts according to NE 03-080, Class II;
- c) EN 9029 - General tolerances for sheet or extruded parts according to NE 03-080, Class III;
- d) EN 9030 - General tolerances for sheet or extruded parts according to NE 03-080, Class IV.

6.1.2 For new drawings the thickness reduction grade shall also be called on drawing through the following Notes (EN), in addition to the geometric tolerance classes as applicable.

- a) EN 9008 - Maximum thickness reduction according to NE 03-080, Grade 9.
- b) EN 9009 - Maximum thickness reduction according to NE 03-080, Grade 13.
- c) EN 9010 - Maximum thickness reduction according to NE 03-080, Grade 17.
- d) EN 9011 - Maximum thickness reduction according to NE 03-080, Grade 23.
- e) EN 9012 - Maximum thickness reduction according to NE 03-080, Grade 30.
- f) EN 9013 - Maximum thickness reduction according to NE 03-080, Grade 40.
- g) EN 9014 - No thickness reduction control regarding forming process according to NE 03-080.

NOTES

- 1) The new drawings shall call both notes, one for the geometric tolerance class and the other for thickness reduction.
- 2) Revision drawings and derived drawings (evolution drawings and/or dash number roll) may maintain the old drawing notes (NI 973/NI 974/NI 975/NI 976) or may use the new ones when a new revision or evolution is released, according to DIP decision.

6.1.3 For new drawings, whenever the use of general tolerances described in Item 5.1.2 is applicable, the note EN 9053 shall also be called on drawing.

EN 9053 - Apply General Tolerances for Holes Located in Flat Web of Straight Contour Flanged Sheet Metal Parts or Not Flanged Flat Sheet Metal Parts, according to NE 03-080.

6.1.4 If more than one surface class or thickness grade are applicable to the same part, besides indicating all applicable ENs in the Part List, each applicable EN shall be indicated inside a flag on the Illustrative Drawing along the respective surface (see Figures 13, 14 and 15).

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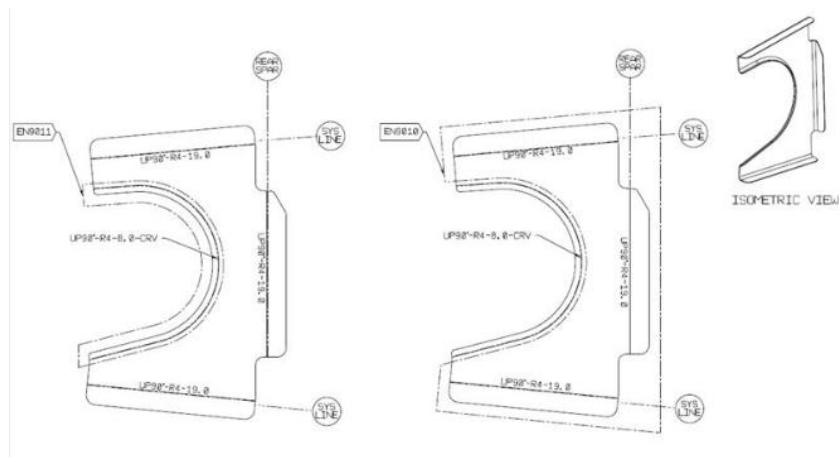


FIGURE 13 : EXAMPLE OF INDICATION OF MORE THAN ONE GRADE IN A STANDARDIZED DETAIL (2D DRAWING)

**Part List -
Example**

Note Code	Description
EN9010	MAXIMUM THICKNESS REDUCTION ACCORDING TO NE03-080 GRADE 17
EN9011	MAXIMUM THICKNESS REDUCTION ACCORDING TO NE03-080 GRADE 23
EN9030	GENERAL TOLERANCES FOR SHEET OR EXTRUDED PARTS ACCORDING TO NE03-080, CLASS IV

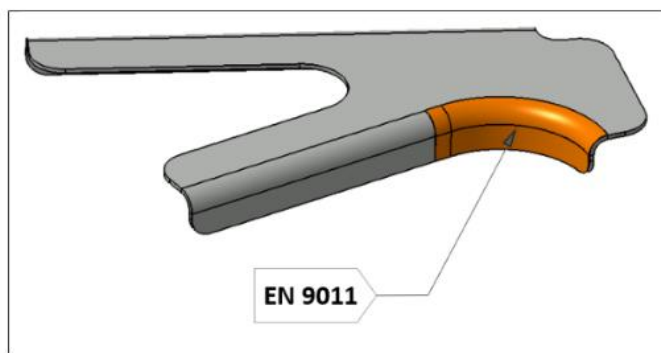
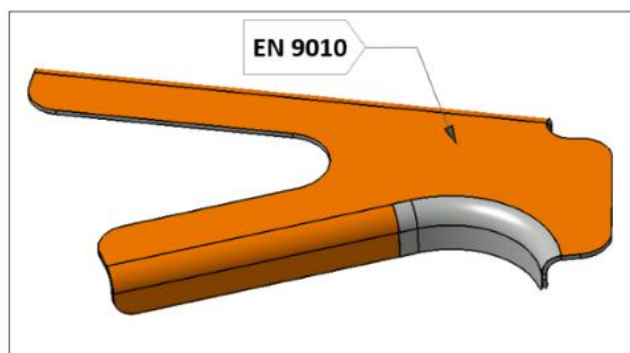


FIGURE 14 : EXAMPLE OF INDICATION OF ONE CLASS AND MORE THAN ONE GRADE IN A STANDARDIZED DETAIL (MBD REPRESENTATION - 3D DRAWING)

**Part List -
Example**

Note Code	Description
EN9014	EN 9014 - NO THICKNESS REDUCTION CONTROL OR INSPECTION
EN9011	MAXIMUM THICKNESS REDUCTION ACCORDING TO NE03-080 GRADE 23
EN9028	GENERAL TOLERANCES FOR SHEET OR EXTRUDED PARTS ACCORDING TO NE03-080, CLASS II

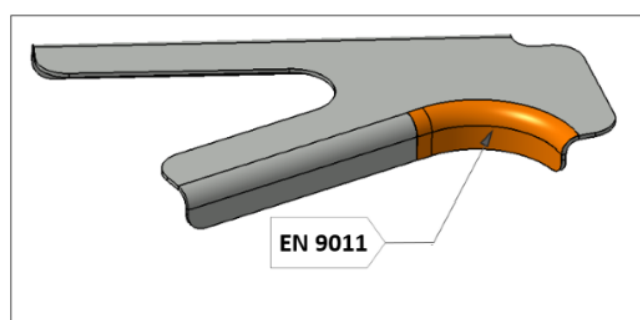
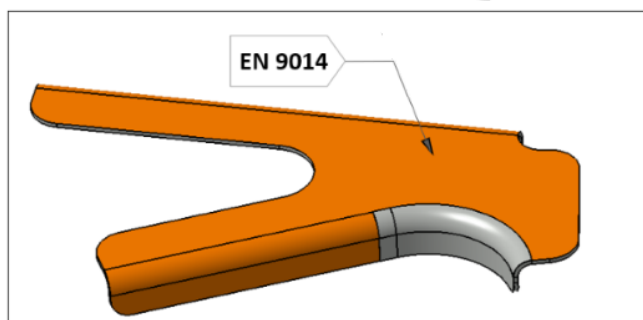


FIGURE 15 : EXAMPLE OF INDICATION OF ONE CLASS AND NO THICKNESS REDUCTION CONTROL AND THICKNESS REDUCTION CONTROL GRADE 23 IN A STANDARDIZED DETAIL (MBD REPRESENTATION - 3D DRAWING)

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6.1.5 If the same grade or the same surface class is applied for all the surfaces of the part, it is not necessary to indicate the respective notes on the illustrative drawing.

6.1.6 Surface class for standardized details such as relief holes, reinforcing channels, etc., shall not be indicated. The standard that standardizes the detail ensures the surface quality (see Figure 13 (NI 457)).

6.1.7 Indicate, if applicable, the "NI 383 - Grain direction not indicated is optional".

6.1.8 If the standard values of general tolerance (0,8 mm) and/or weight (0,5 kgf or 1,0 kgf) are not adequate to check warping (see Table 6), the following Item Notes (NI) shall be called out on drawing:

NI 977: Maximum allowable warping <*> mm and warping control procedures according to NE 03-080.

<*> indicate: a "specific value" for warping tolerance or "NA" (Not Applicable) for parts that do not need warping control.

If there is more than one specific value for warping tolerance, indicate the first one using NI 977 and the others using Free Notes (NL) with the same writing as NI 977.

NI 978: Maximum weight application <*> kgf at the indicated points and warping control procedures according to NE 03-080.

<*> indicate the maximum weight value.

If there is more than one "weight value", indicate the first one using NI 978 and the others using Free Notes (NL) with the same writing as NI 978.

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